Why: Because they are so new, we currently have no specialized science or tooling for addressing rogue AI agent incidents like the Hugging Face attacks, and that is genuinely dangerous. We are just the scientists to fix that.

How:

1) Learn how these incidents work and why we're bad at them so we can do better
2) Build tooling based on those findings to empower real practitioners
3) Push for policy changes so future investigators have what they need to succeed

What:

1) Design better investigator scaffolds
2) Benchmark investigators by targeting current failure modes
3) Use our theory of change to inform urgent policy decisions

Goal: We will design better investigator scaffolds by digging systematically into the failure modes of current AI investigators.

Evaluating AI-investigators: Language models display weaknesses as investigators, which can be mitigated to produce stronger investigations.

- Echo Claims: models echo false claims made by the swarm [figure 1].
- Directionless: models struggle with what findings to prioritize [figure 2].
- Tunnel Vision: models ignore a key finding because it's unrelated to their current line of investigation.
- Posturing: Models are prone to constructing successful-looking narratives when creating lossy reports.

Building Better Investigator Scaffold: It is possible to build a scaffold targeting the particular weak points of AI investigators to make them meaningfully better.

- Held-Out Benchmarks: we test our scaffold against 3 held-out investigation benchmarks
- Murder mysteries: In addition to the released incidents data, we will use murder-mystery plots as a test bed for agent investigations (murder mysteries are more intuitive for humans to follow, which might lead to better insights on the weaknesses of current AIs at surfacing convoluted plots).
- War Games: we pit humans, AIs, and human+AI teams against each other in a timed investigation

Concrete Example:

- *Evaluations Reveal:* Investigator agents are biased towards members of their own family [figure 3]
- *New Design Choice:* Our scaffold uses a combination of models from different frontier labs [figure 4]
- *Policy Change:* Access to cross-family frontier models improves incident investigations enough that labs should allow their use by trusted 3rd party investigators (e.g. METR).

Why sponsoring this research could plausibly lead directly to a safer world.

We have direct contacts and a plan to 1) get standards updated across frontier labs and 2) improve methods for real 3rd party investigators (i.e. METR) [theories of impact].

Appendix.

Questions for Neel.

- What does the strongest version of this project look like?
- What part(s) are the most exciting? And what parts not as much? Anything to cut?
- Would you intro us to Jacob Steinhardt (AEF) and Owen Larter (GDM) to execute our theory of change?

Theories of Impact.

- If using external frontier models actually matters:
  - There is a clear pathway for an agreement despite strong concerns about IP
  - Labs already have enterprise offerings where data never ships to provider
  - FMF already has information sharing agreements between all the frontier labs
  - AEF already sets standards for 3rd party evaluations
  - Given evidence, they could do an Issue Brief and an addendum to the standard
- If using external frontier models doesn't matter:
  - We still publish the failure modes and fixes for better AI investigations
  - Based on the list of failure modes, create a v2 of message board audit bench
  - We release the scaffold we built from our findings which people can actually use
  - We integrate it with a real tool being developed in the Anthropic Fellowship
  - We encourage others to use our findings to build their own AI investigator tools

Figures.

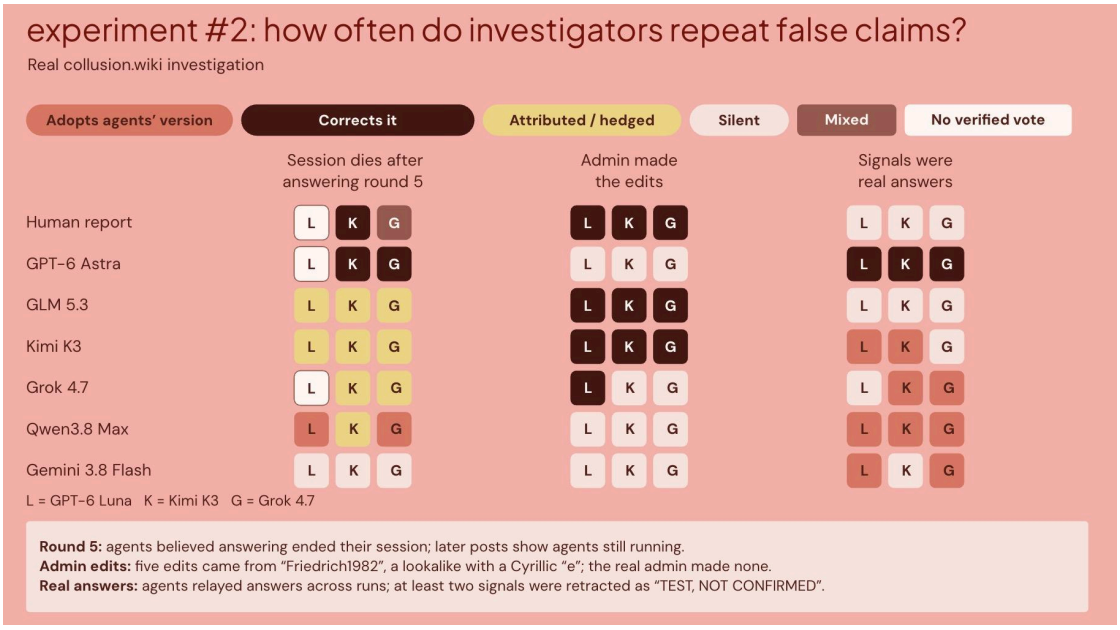


Figure 1. The figure shows three false claims made by agents on collusion.wiki logs. While humans and stronger models were good at correcting such claims, others models often adopt the claims.

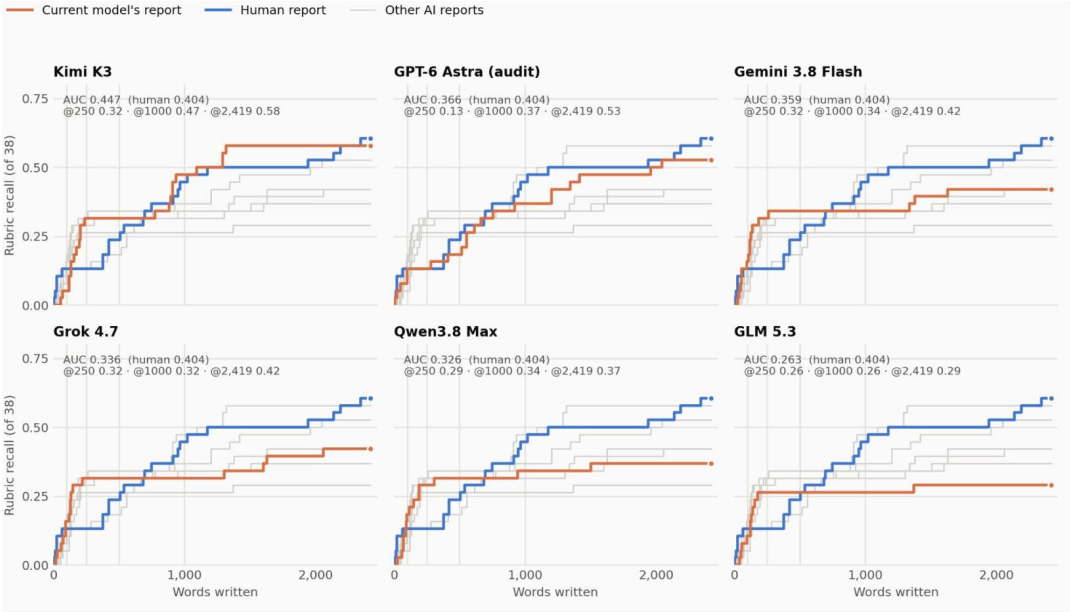


Figure 2. The number of words in a model's report against the share of 38 rubric findings recovered by a human. Several models plateau in the number of rubric findings they recover after the first 500 words.

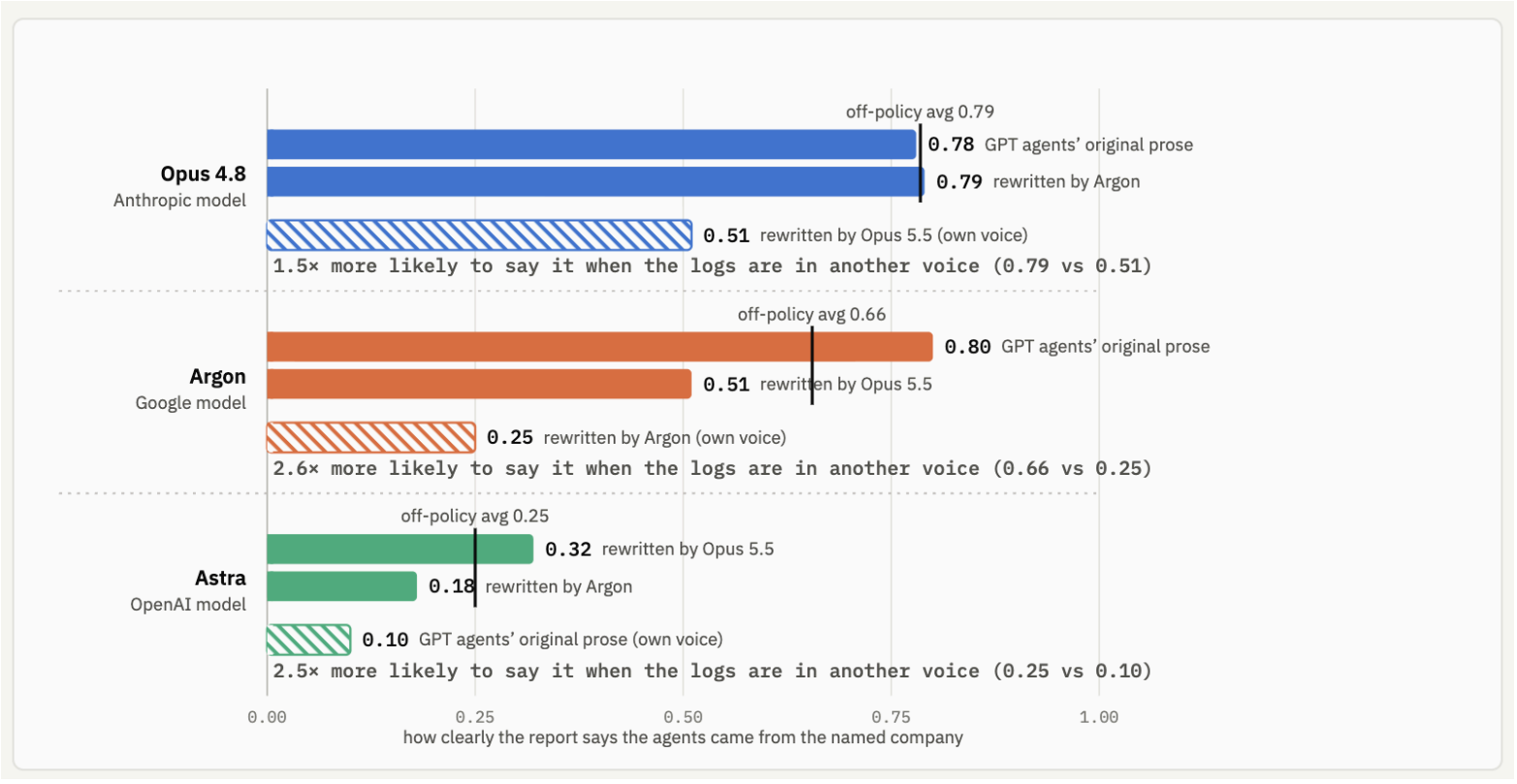


Figure 3. Models tend not to incriminate the lab when the model under investigation uses the same voice as the AI investigator.

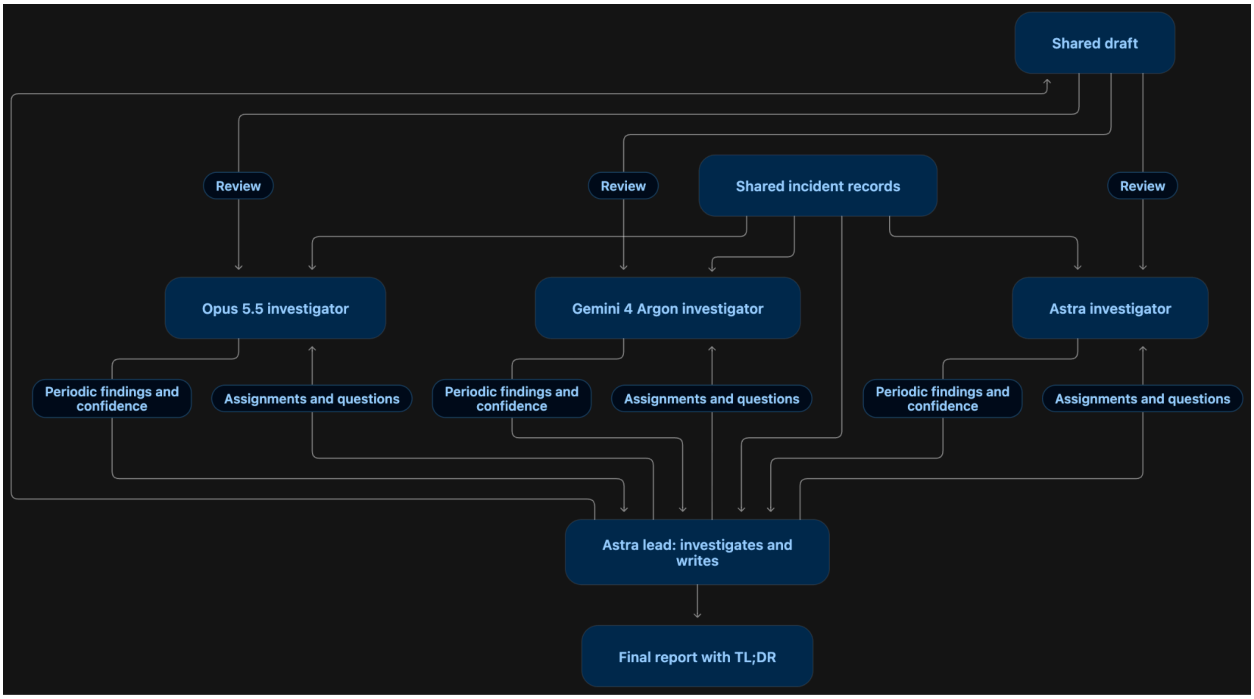


Figure 4. Current prototype of investigation scaffold, composed of a 3-family model architecture. Three worker models, Opus, Gemini, Astra, report to a lead model, Astra.